Iteration 1

$y = mx + b$ / $y = m_1x_1 + m_2x_2 + b$

$$x = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \quad y = \begin{bmatrix} 5 \\ 6 \end{bmatrix} \quad m = \begin{bmatrix} -1 \\ 2 \end{bmatrix} \quad b = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Learning rate used $= 0.01 = \alpha$

$e = \hat{y} - y$

$$= \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

$MSE / d = \frac{1}{n} \sum e^2$

$$= \frac{1^2 + 11^2}{2} \longrightarrow \frac{1 + 121}{2} \longrightarrow = \underline{61}$$

Derivation

$$d = \frac{1}{n}\sum e^2$$

$$\frac{dd}{dm} = \frac{1}{n}(2e_1x_{11} + 2e_2x_{21})$$

$$= \frac{2}{n}(x_1e_1 + x_2e_2)$$

$$\frac{dd}{db} = \frac{1}{n}(2ex1)$$

$$= \frac{2}{n}e$$

$$\alpha\begin{bmatrix} dd/dm1 \\ dd/dm_2 \end{bmatrix} = \frac{2}{n}\begin{bmatrix} x_{11} & x_{21} \\ x_{12} & x_{22} \end{bmatrix}\begin{bmatrix} e_1 \\ e_2 \end{bmatrix}$$

$$\frac{dd}{dm} = \frac{2}{n}X^Te$$

$$X^T = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix}$$

$$\frac{dd}{dm} = \frac{2}{2}\begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix}\begin{bmatrix} 1 \\ 11 \end{bmatrix} = \begin{bmatrix} 1X1 + 4X11 \\ 3X1 + 10X11 \end{bmatrix}$$

$$= \begin{bmatrix} 1 + 44 \\ 3 + 110 \end{bmatrix} = \begin{bmatrix} 45 \\ 113 \end{bmatrix}$$

$$\frac{dd}{dm} = \begin{bmatrix} 45 \\ 113 \end{bmatrix}$$

$$\frac{dd}{db} = 2 \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

$$\frac{dd}{db} = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

$$\text{new } m = \text{old } m - \alpha \frac{dd}{dm}$$

$$= \begin{bmatrix} -1 \\ 2 \end{bmatrix} - 0.01 \begin{bmatrix} 45 \\ 113 \end{bmatrix} = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix}$$

$$\text{new } b = \text{old } b - \alpha \frac{dd}{db}$$

$$= \begin{bmatrix} 1 \\ 1 \end{bmatrix} - 0.01 \begin{bmatrix} 1 \\ 11 \end{bmatrix} = \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix}$$

$$\text{new } m = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix} \qquad \text{new } b = \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix}$$

Iteration 1 = $m = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix}^T$, $b = \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix}^T$

Iteration 2

$$X_m = \begin{bmatrix} 1 \cdot -1.45 + 3 \cdot 0.87 \\ 4 \cdot -1.45 + 10 \cdot 0.87 \end{bmatrix} = \begin{bmatrix} -1.45 + 2.61 \\ -5.80 + 8.70 \end{bmatrix}$$

$$= \begin{bmatrix} -1.45 + 2.61 \\ -5.80 + 8.70 \end{bmatrix} = \begin{bmatrix} 1.16 \\ 2.90 \end{bmatrix}$$

$$\hat{y} = \begin{bmatrix} 1.16 + 0.99 \\ 2.90 + 0.89 \end{bmatrix} = \begin{bmatrix} 2.15 \\ 3.79 \end{bmatrix}$$

$$e = \begin{bmatrix} 2.15 - 5 \\ 3.79 - 6 \end{bmatrix} = \begin{bmatrix} -2.85 \\ -2.21 \end{bmatrix}$$

$$J = \frac{1}{2}\left((-2.85)^2 + (-2.21)^2\right) = \frac{1}{2}(8.1225 + 4.8841)$$

$$= \frac{1}{2}(13.0066) = 6.5033$$

CS Scanned with CamScanner

$$X^T e = \begin{bmatrix} 1 \cdot -2.85 + 4 \cdot -2.21 \\ 5 \cdot -2.85 + 10 \cdot -2.21 \end{bmatrix} = \begin{bmatrix} -2.85 - 8.84 \\ -8.55 - 22.10 \end{bmatrix}$$

$$= \begin{bmatrix} -11.69 \\ -30.65 \end{bmatrix}$$

$$\frac{\partial J}{\partial m} = 1 \times \begin{bmatrix} -11.69 \\ -30.65 \end{bmatrix} = \begin{bmatrix} -11.69 \\ -30.65 \end{bmatrix}$$

$$\frac{\partial J}{\partial b} = 1 \times \begin{bmatrix} -2.85 \\ -2.21 \end{bmatrix} = \begin{bmatrix} -2.85 \\ -2.21 \end{bmatrix}$$

$\alpha = 0.01$

$$\frac{\alpha \partial J}{\partial m} = \begin{bmatrix} -0.1169 \\ -0.3065 \end{bmatrix}, \quad \frac{\alpha \partial J}{\partial b} = \begin{bmatrix} -0.0285 \\ -0.0221 \end{bmatrix}$$

$$m = \begin{bmatrix} -1.45 - (-0.1169) \\ 0.87 - (-0.3065) \end{bmatrix} = \begin{bmatrix} -1.45 + 0.1169 \\ 0.87 + 0.3065 \end{bmatrix}$$

$$= \begin{bmatrix} -1.3331 \\ 1.1765 \end{bmatrix}$$

Scanned with CamScanner

$$b = \begin{bmatrix} 0.99 - (-0.0285) \\ 0.89 - (-0.0221) \end{bmatrix} = \begin{bmatrix} 0.99 + 0.0285 \\ 0.89 + 0.0221 \end{bmatrix}$$

$$= \begin{bmatrix} 1.0185 \\ 0.9121 \end{bmatrix}$$

Iteration 2 = $M = \begin{bmatrix} -1.3351 \\ 1.1765 \end{bmatrix}^T$ $b = \begin{bmatrix} 1.0185 \\ 0.9121 \end{bmatrix}^T$

Iteration 3

$$X_m = \begin{bmatrix} 1 \cdot -1.3331 + 3 \cdot 1.1765 \\ 4 \cdot -1.3331 + 4 \cdot 1.1765 \end{bmatrix} = \begin{bmatrix} -1.3331 + 3.5295 \\ -5.3324 + 4.7060 \end{bmatrix}$$

$$= \begin{bmatrix} 2.1964 \\ 6.4326 \end{bmatrix}$$

$$\hat{y} = \begin{bmatrix} 2.1964 + 1.0185 \\ 6.4326 + 0.9121 \end{bmatrix} = \begin{bmatrix} 3.2149 \\ 7.3447 \end{bmatrix}$$

Scanned with CamScanner

Formative 3 Update 3

$m = \begin{bmatrix} -1.3331 \\ 1.1765 \end{bmatrix}$, $b = \begin{bmatrix} 1.0185 \\ 0.9121 \end{bmatrix}$, $x = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}$, $y = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$, $n = 2$

$\hat{y} = xm + b = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}\begin{bmatrix} -1.3331 \\ 1.1765 \end{bmatrix} + \begin{bmatrix} 1.0185 \\ 0.9127 \end{bmatrix}$

$\hat{y} = \begin{bmatrix} (-1.3331)(1) + (1.1765)(3) \\ (-1.3331)(4) + (1.1765)(10) \end{bmatrix} + \begin{bmatrix} 1.0185 \\ 0.9121 \end{bmatrix} = \begin{bmatrix} 2.1964 \\ 6.4326 \end{bmatrix} + \begin{bmatrix} 1.0185 \\ 0.9121 \end{bmatrix} =$

$\hat{y} = \begin{bmatrix} 3.2149 \\ 7.3447 \end{bmatrix}$

error $= \hat{y} - y$

$\hat{y} - y = \begin{bmatrix} 3.2149 \\ 7.3447 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} -1.7851 \\ 1.3447 \end{bmatrix}$

$MSE = \frac{1}{2}\left[(-1.7851)^2 + (1.3447)^2\right] = 2.4974 = \text{cost } J$

Gradients... dj/dm_1, dj/dm_2, dj/db

$J = \frac{1}{n}\sum_{i=1}^{n} (\hat{y} - y)^2$, but $\hat{y} = m_1x_1 + m_2x_2 + b$

so $J = \frac{1}{n}\sum_{i=1}^{n} ((m_1x_1 + m_2x_2 + b) - y)^2$ | Also, $e = \hat{y} - y$, $e = (m_1x_1 + m_2x_2 + b) - y$ | so $J = \frac{1}{n}\sum_{i=1}^{n} e^2$

$\frac{dJ}{de} = \frac{2}{n}\sum_{i=1}^{n} e$, $\frac{dJ}{dm_1} = \frac{dJ}{de} \cdot \frac{de}{dm}$ | $\frac{de}{dm_1} = x_1 + 0 + 0 - 0 = x_1$

so $\frac{dJ}{dm_1} = \frac{2}{n}\sum_{i=1}^{n} (\hat{y} - y)x^1$. same for m_2

$\frac{dj}{dm_1} = \frac{2}{2}\left[\begin{bmatrix} -1.7851 \\ 1.3447 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 4 \end{bmatrix}\right] = 3.5937$ | $\frac{dJ}{dm_2} = \frac{2}{2}\left[\begin{bmatrix} -1.7851 \\ 1.3447 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 10 \end{bmatrix}\right] = 8.0917$

$\frac{dJ}{db} = \frac{2}{2}\begin{bmatrix} -1.7851 \\ 1.3447 \end{bmatrix} = \begin{bmatrix} -1.7851 \\ 1.3447 \end{bmatrix}$

$m_{1new} = m_{1old} - a\frac{dJ}{dm_1} = -1.3331 - 3.5937a$

$m_{2new} = m_{2old} - a\frac{dJ}{dm_2} = -1.1765 - 8.0917a$

$b_{new} = b_{old} - a\frac{dJ}{db} = \begin{bmatrix} 1.0185 \\ 0.9121 \end{bmatrix} - a\begin{bmatrix} -1.7851 \\ 1.3447 \end{bmatrix}$

Continuation of Update 3...

Let $a = 0.01$

$m_{1,new} = -1.3331 - 3.5937(0.01) = -1.3690$

$m_{2\,new} = -1.765 - 8.0917(0.01) = 1.0956$

$b_{new} = \begin{bmatrix} 1.0185 \\ 0.9121 \end{bmatrix} - \begin{bmatrix} -0.0179 \\ 0.0134 \end{bmatrix} = \begin{bmatrix} 1.0364 \\ 0.8937 \end{bmatrix}$

New MSE, (cost J)

$J = \frac{1}{n} \sum_{i=1}^{n} (\hat{y} - y)^2 \quad | \quad \hat{y} = \begin{bmatrix} -1.3690 + 3.2868 \\ -5.4760 + 10.9560 \end{bmatrix} + \begin{bmatrix} 1.0364 \\ 0.8937 \end{bmatrix} \begin{bmatrix} 2.594 \\ 6.3787 \end{bmatrix}$

$\hat{y} - y = \begin{bmatrix} -2.0458 \\ 0.3887 \end{bmatrix}$ $MSE = \frac{1}{2} (4.1853 + 0.1434) = 2.1644$

iteration	m_1	m_2	b_1 b_2	MSE
0	-1.0000	2.0000	[1.0000, 1.0000]	61.0000
1	-1.4500	0.8700	[0.9900, 0.8900]	6.5033
2	-1.3331	1.1765	[1.0185, 0.9121]	2.4974
3	-1.3690	1.0956	[1.0364, 0.8987]	2.1645

Conclusion

m and b are observed to be moving in a direction that reduces the error as seen in MSE value dropping from 61 to 2.1645 in the third iteration/update.